Predicting Hindi Speech Duration from Text

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Machine Learning and Artificial Intelligence (MAP 1602)

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Problem Formulation

Machine Learning Objective:

- Predict how long it takes to speak Hindi text

Why Machine Learning:

- Complex relationship between text and speech rate
- Not simply character count
- Depends on linguistic features

ML TASK TYPE : REGRESSION

Input Features (X):

- **Characters :** The raw length of the text.
- **Matra** :The count of vowel modifiers.
- **Halant :** The count of vowel suppression markers

Example :

शक्ति = श + क + ् + त + ि = 5

5 Characters, 1 Matra, 1 Halant

Data Preparation

1. Dataset:

- 4,586 audio files with transcripts

2. Features Extracted:

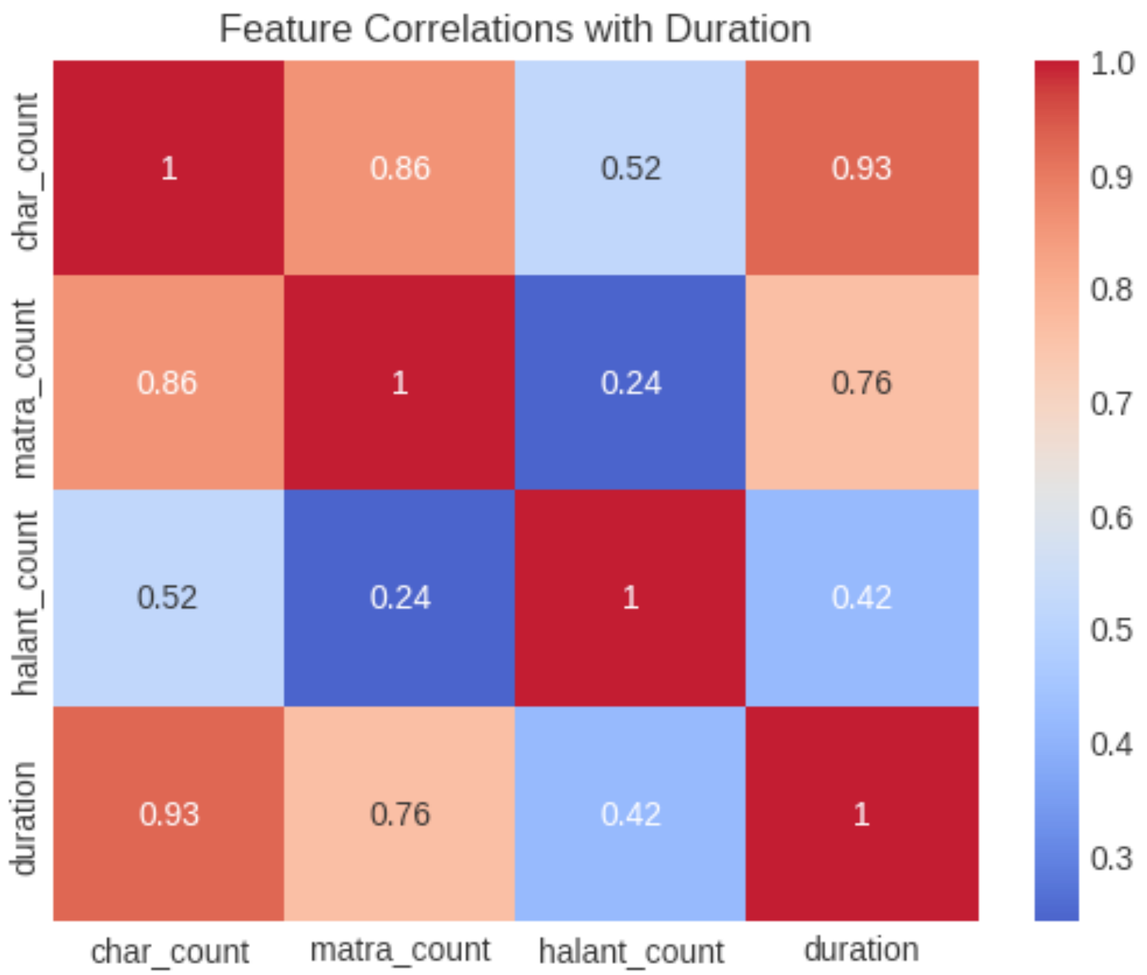
- **char_count** → Mean: 57.54, Range: 4-115
- **matra_count** → Mean: 15.17, Range: 0-36
- **halant_count** → Mean: 1.86, Range: 0-11
- **duration** → Mean: 5.34s, Range: 0.95-12.88s

3. Train-Test Split:

- 80-20 (3,669 train, 917 test)

4. Correlations with Duration:

- char_count : 0.93 (strong linear)
- matra_count : 0.76 (moderate-strong)
- halant_count : 0.42 (moderate)

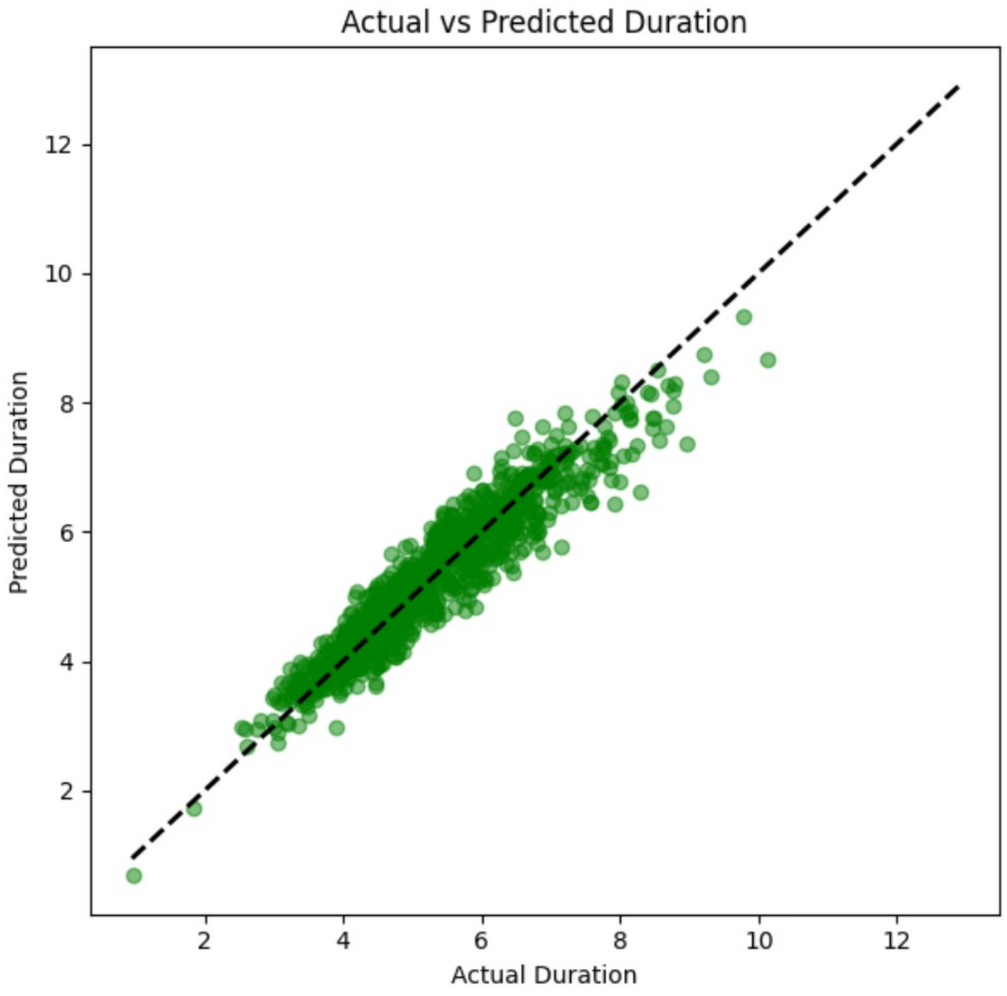


Baseline Model

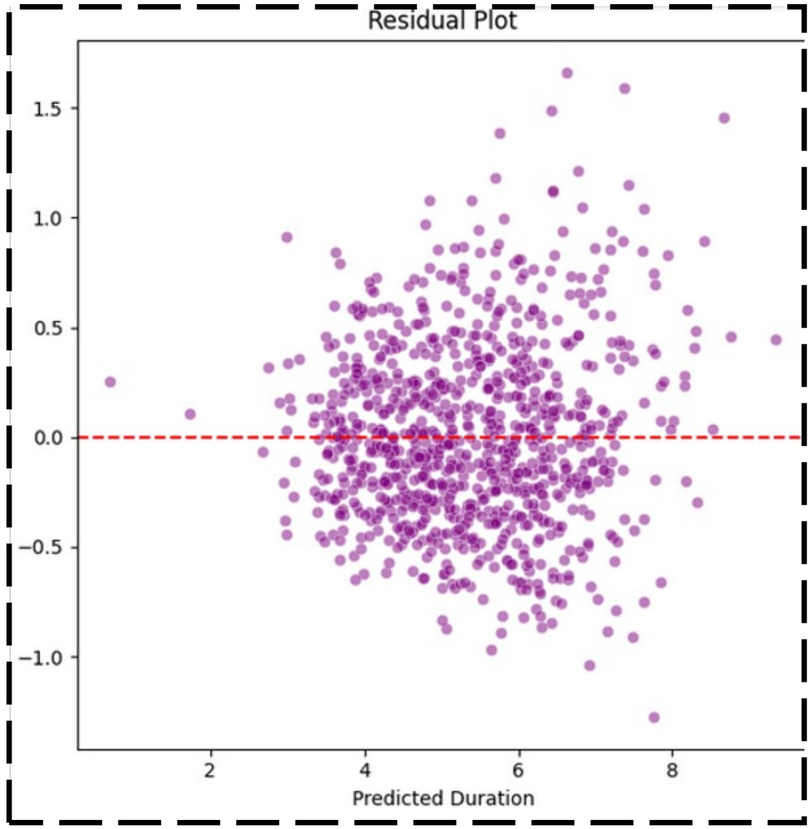
Why Linear Regression:

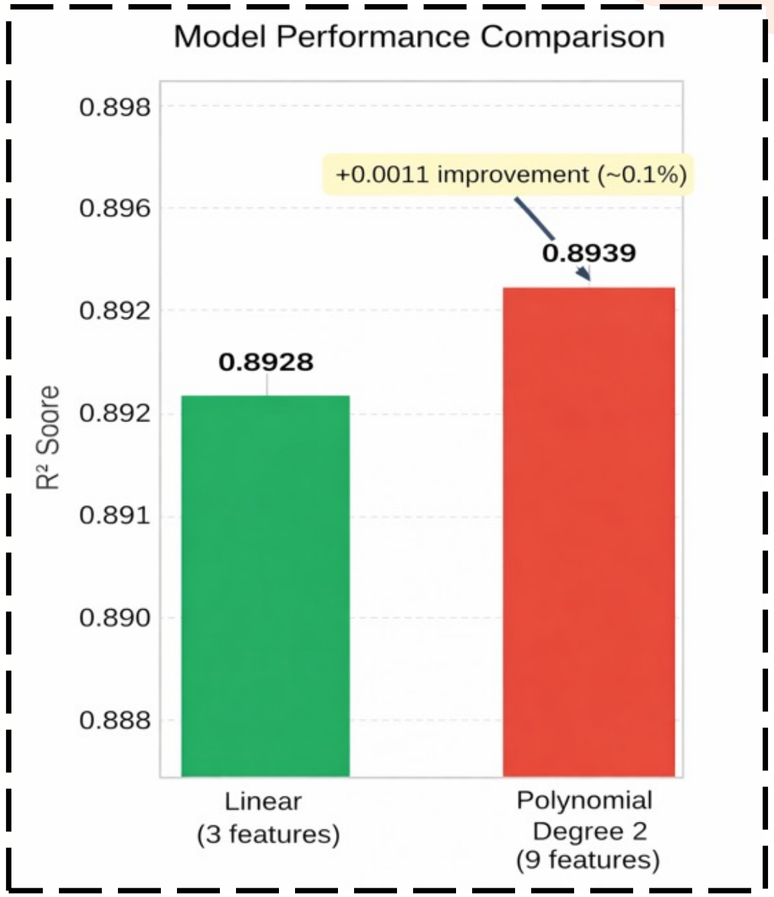
- Simple and interpretable
- Strong correlations suggest linear relationship
- Provides coefficient for each feature

Performance Metrics		
MAE	RMSE	R2 Score
0.3624	0.4148	0.8928



Model Evaluation



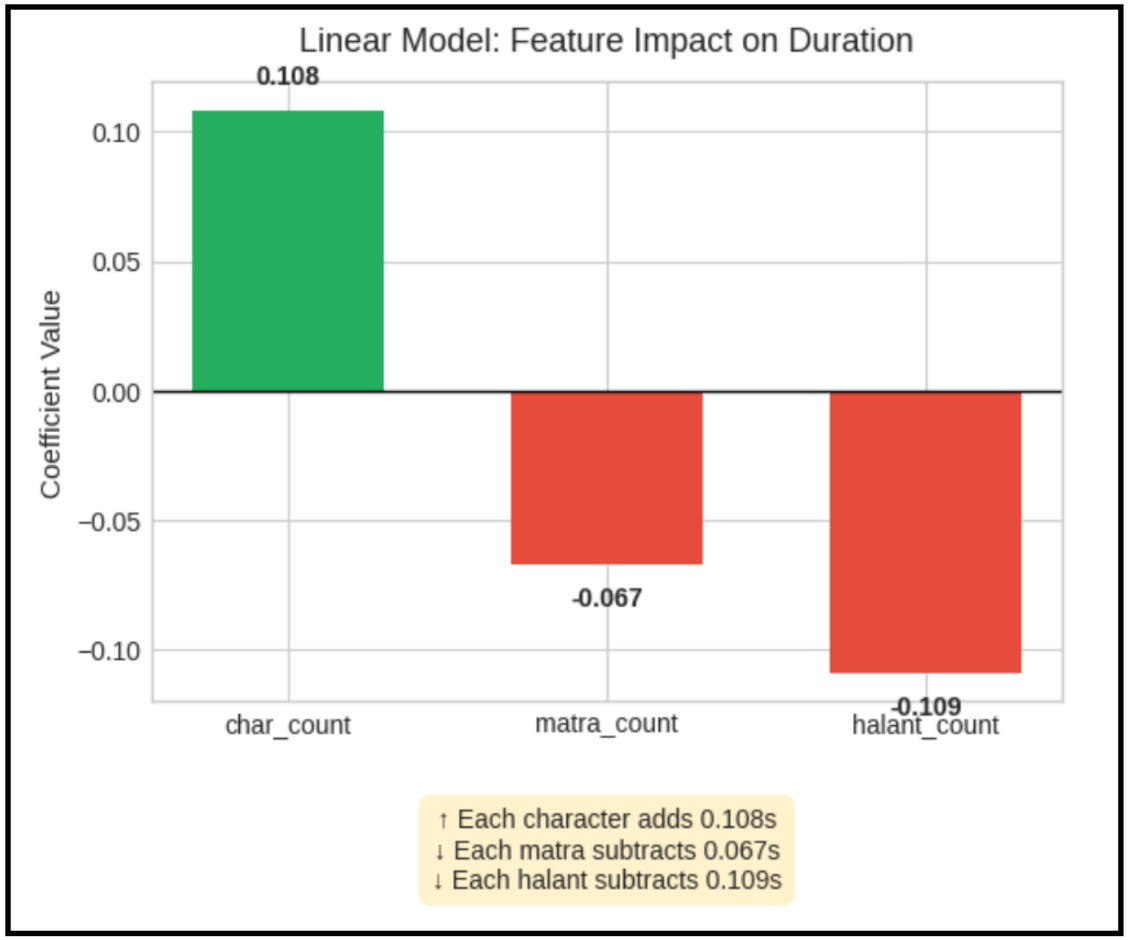


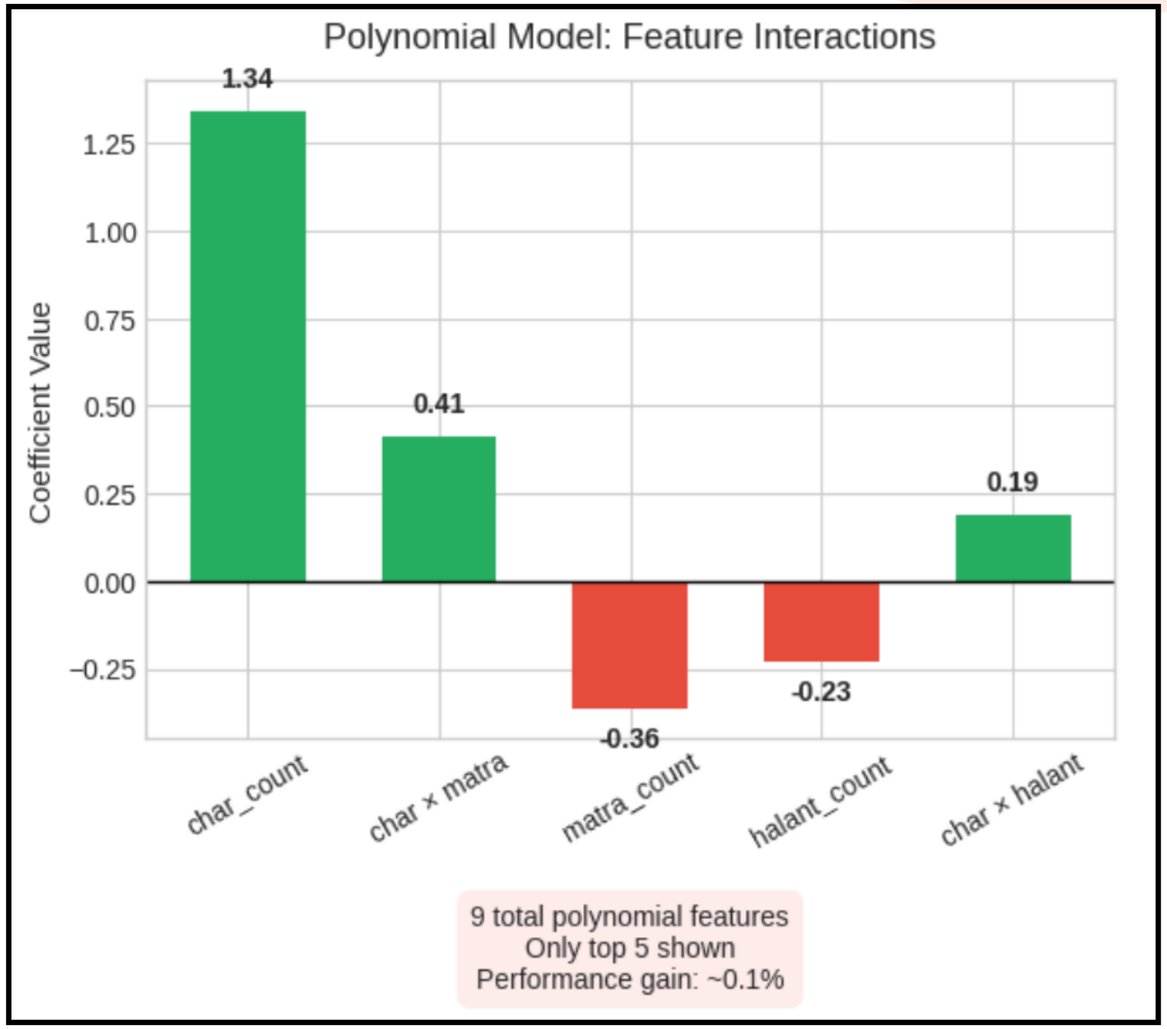
Residual Analysis:

- Mean residual: 0.0116 (≈ zero - no bias)
- Random scatter pattern (linearity holds)
- Normal distribution confirmed

Linear Model is Optimal.

Feature Importance / Interpretability





Results & Insights

What We Discovered:

1.Linear model explains 89% of variance

- Simple features capture most patterns

2. The "Efficiency Paradox":

- Example: 4 chars & 0 halants → 0.77s (0.19s/char),
  5 chars & 1 halant → 0.70s (0.14s/char)
- Adding a halant REDUCES per-char time by 26%!

3. Linguistic Interpretation:

- Hindi speakers articulate conjuncts efficiently

4. Matras and halants create efficiency in speech

- Fluent speakers handle complexity smoothly

Limitations & Future Work

Current Limitations	Future Improvements:
1. Single speaker only. (needs validation)	1.Collect multi-speaker data
2. Studio quality. (No real-world noise)	2. Real-world applications: Audiobook pacing , Language learning apps , Public speaking coaches
3. Read speech only (no conversation)	3. Add more features (position, context)

Workflow Summary

Data Preparation

- Feature Extraction

Model Training

- Train Test
- Linear Regression

Results

- Evaluation

Google Collab Link :
https://colab.research.google.com/drive/1XgI4PsBQ61nGAVx7-Dbkcb06rE6qeT11?usp=sharing